Name: _____

WORKSHEET

Translating exponential graphs

1 Find the y-intercept of the following functions

a $y = 7 \times 3^x$ _____

b $f(x) = -2 \times 10^x$ _____

c $f(p) = 0.8 \times 0.1^p$ _____

d $f(t) = 0.125^t - 0.375$ _____

e $y = 4^x + 3$ _____

f $y = 10 + 4 \times 2000^x$ _____

g $f(z) = 3^{x+2}$ _____

h $y = 3 \times 2^{x+4} + 5$ _____

2 Sketch the following graphs and label the y-intercept.

a $y = 3 \times 2^x$

b $y = 0.5 \times 2^x$

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c $y = -3 \times 2^x$

d $y = 3 \times 2^{-x}$

3 Consider the graph of $y = 2^x$. For the graphs in question **2**, describe how they compare to $y = 2^x$.

4 **a** $y = 10^x$

b $y = 0.5^x$

c $y = 2^{-x}$

d $y = -2^{-x}$

e What do you notice about graphs **b** and **c**?

f How does the value of a affect the graph of the exponential $y = a^x$?

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5 Sketch the following graphs

a $y = 2^x - 4$

b $y = 2^x + 1$

c $y = 4 \times 2^x - 4$

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d $y = 5 \times 2^x - 2$

e How does the value of c affect the graph of the exponential $y = a^x + c$?

6 Sketch the following graphs

a $y = 2^{x+2}$

b $y = 2^{x-1}$

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c $y = 2 \times 10^{x+1}$

d $y = 0.5^{x-4}$

e How does the value of b affect the graph of the exponential $y = a^{x+b}$?

7 Sketch the following graphs

a $y = 2^{x+2} - 5$

b $y = 2^{x-1} + 1$

c $y = 0.5 \times 3^{x-2} - 1$

d $y = 4 \times 0.5^{x+3} - 0.5$

Answers

1 **a** 7

b -2

c 0.8

d 0.625

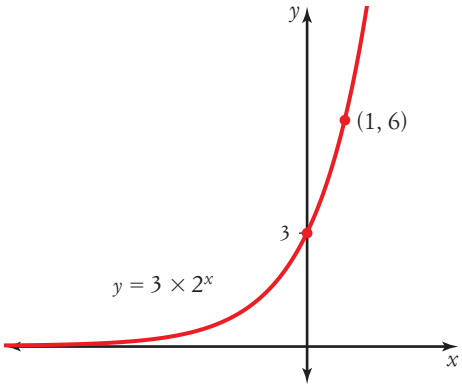
e 4

f 14

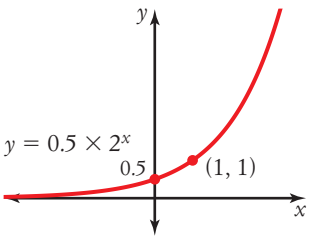
g 9

h 53

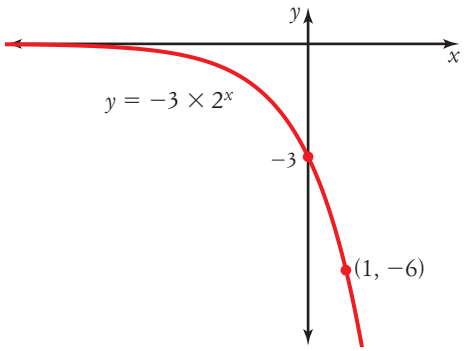
2 **a** $y = 3 \times 2^x$



b $y = 0.5 \times 2^x$

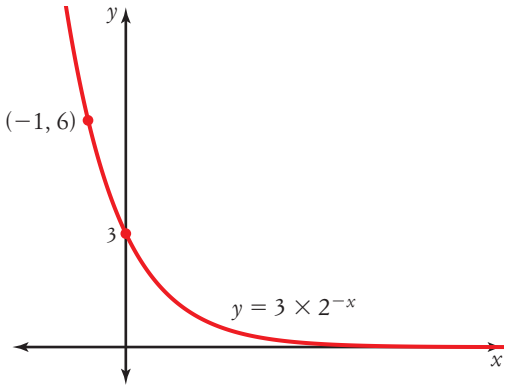


c $y = -3 \times 2^x$

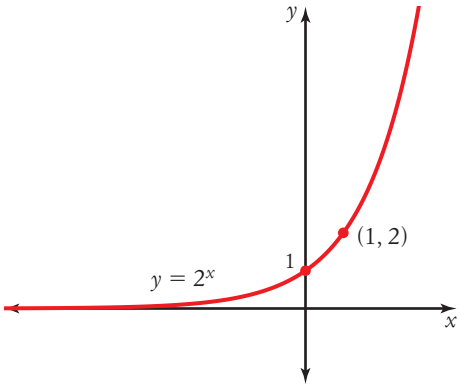


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d $y = 3 \times 2^{-x}$



3 $y = 2^x$



a The y-intercept has shifted to 3 and the graph has been stretched vertically (horizontal asymptote has not shifted vertically)

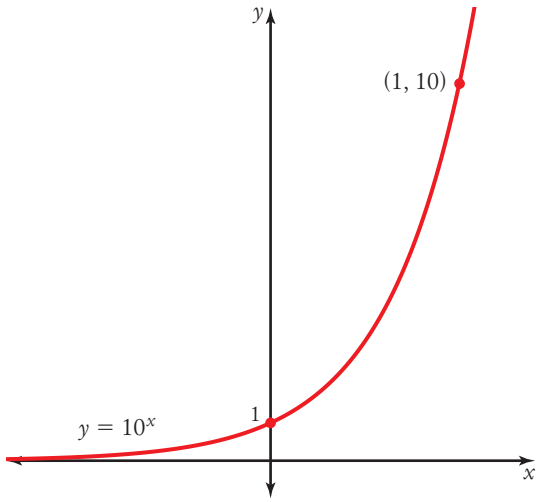
b The y-intercept has shifted to 0.5 and the graph has been compressed vertically

c The graph has been reflected about the x-axis, and the y-intercept has been shifted away from the origin as the graph has been vertically stretched.

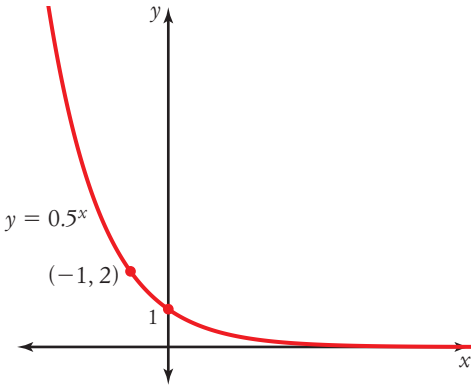
d The graph has been reflected about the y-axis, and the y-intercept has been shifted to 3 as the graph has been vertically stretched.

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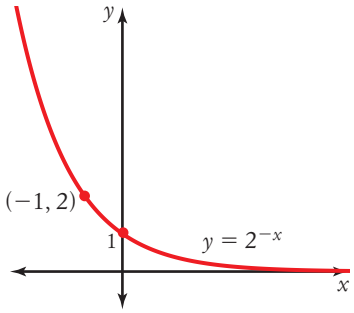
4 a $y = 10^x$



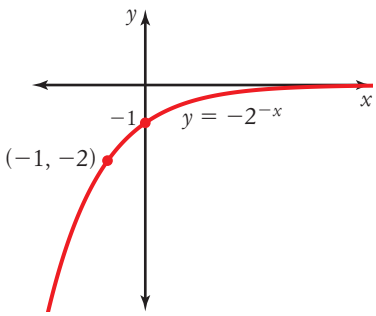
b $y = 0.5^x$



c $y = 2^{-x}$



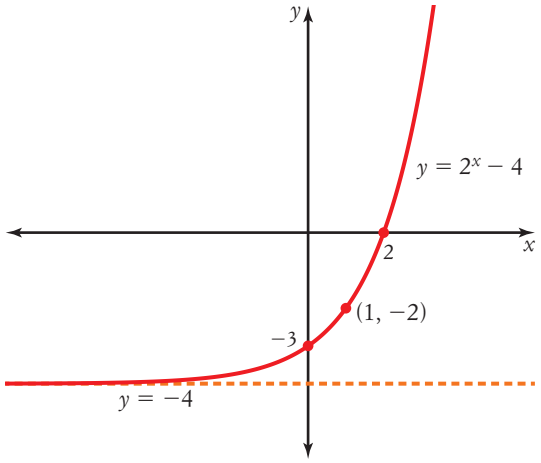
d $y = -2^{-x}$



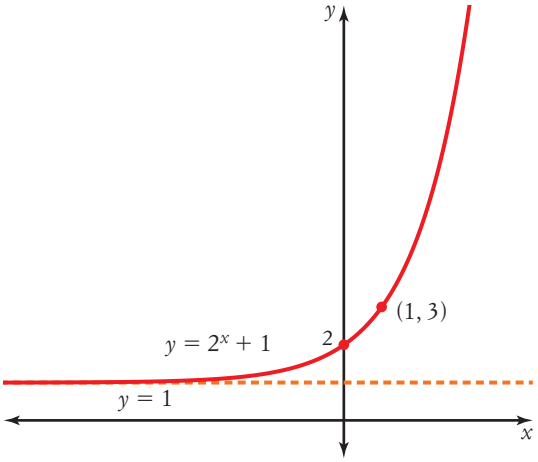
e They are the same graph as $0.5^x = \left(\frac{1}{2}\right)^x = \frac{1^x}{2^x} = \frac{1}{2^x} = 2^{-x}$.

f The exponential is increasing if $a > 1$ and decreasing if $0 < a < 1$. Note that $-2^{-x} = -1 \times 2^{-x}$ so here $a = 2$. If a is large, $a >> 1$, then the growth of the exponential is much faster as x increases.

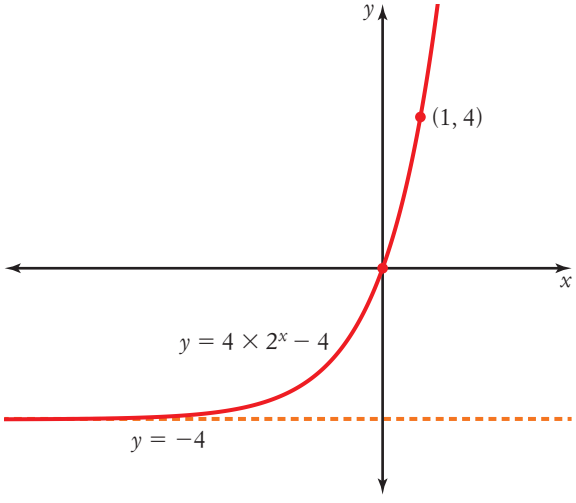
5 a $y = 2^x - 4$



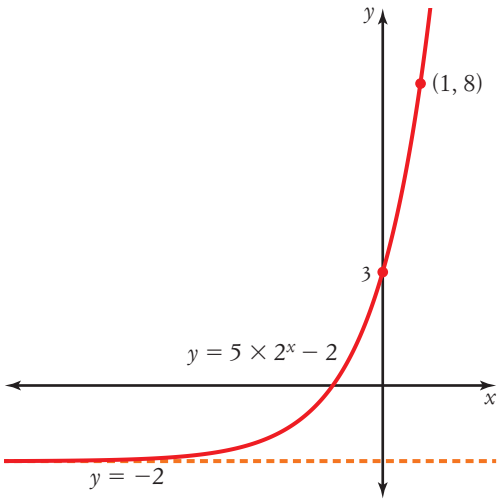
b $y = 2^x + 1$



c $y = 4 \times 2^x - 4$

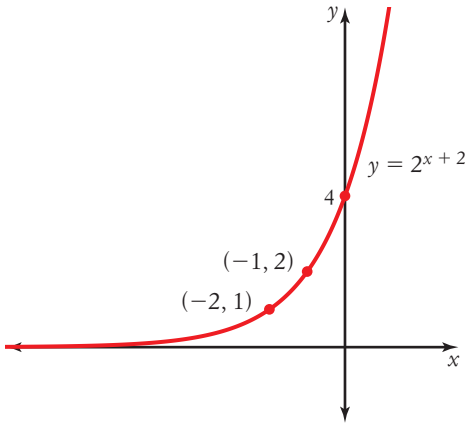


d $y = 5 \times 2^x - 2$

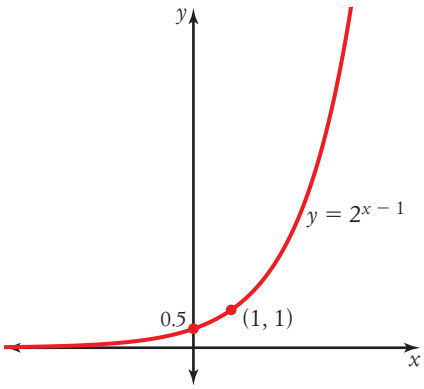


e The exponential including the horizontal asymptote is vertically shifted by the value of c. It is shifted upward if $c > 0$ and downward if $c < 0$.

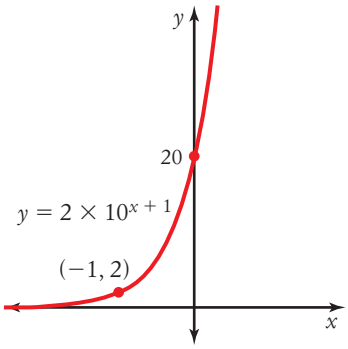
6 a $y = 2^{x+2}$



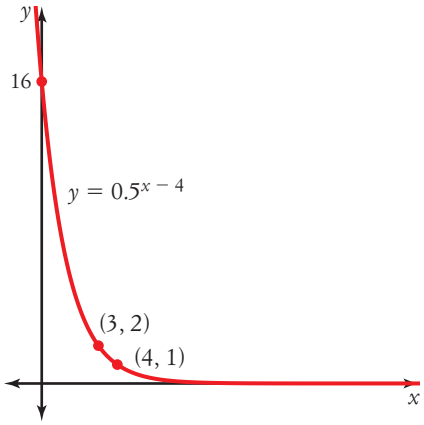
b $y = 2^{x-1}$



c $y = 2 \times 10^{x+1}$

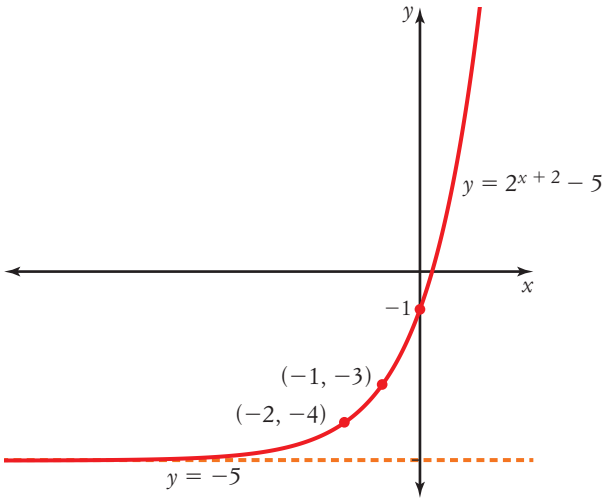


d $y = 0.5^{x-4}$

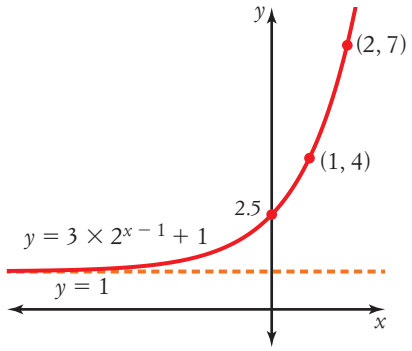


e The exponential is horizontally shifted to the right by an amount b when $b < 0$ and horizontally shifted to the left by an amount b when $b > 0$. The horizontal asymptote remains unchanged.

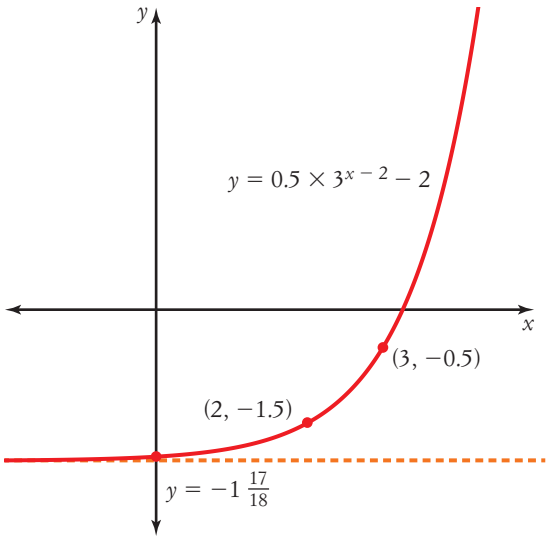
7 a $y = 2^{x+2} - 5$



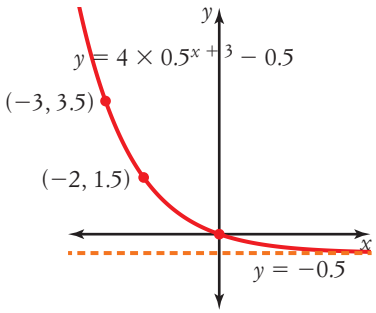
b $y = 3 \times 2^{x-1} + 1$



c $y = 0.5 \times 3^{x-2} - 2$



d $y = 4 \times 0.5^{x+3} - 0.5$



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